

LCI699 (osilodrostat)

CLCI699C2301 / NCT02180217

A Phase III, multi-center, double-blind, randomized withdrawal study of LCI699 following a 24 week, single-arm, open-label dose titration and treatment period to evaluate the safety and efficacy of LCI699 for the treatment of patients with Cushing's disease

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Document History – Changes compared to previous final version of SAP

Date	Time point	Reason for update	Outcome for update	Section and title impacted (Current)
08-Apr-2015	Prior to DB lock	Creation of final version	N/A - First version	NA
05-Dec-2017	Prior to DB lock	Creation of SAP Amendment 1	Change from RAP module 3 to new SAP template. Changes made to align with study Protocol Amendment 4 Clarifying how the proportion of responder will be calculated for timepoints in extension. Providing criteria of adequate follow-up for time points in the extension period Subgroup analysis added Changes related to PRO section Changes made following team review of TFL shell. Edits made to correct typo or improve clarity/consistence of the document.	Throughout the SAP, the sections from the previous RAP M3 have been mapped to the relevant section in the SAP template. To align with study protocol amendment 4, following changes were made: Section 1: end of extension is changed from 'Week 96' to 'Week 72' Section 1.1: added information regarding protocol amendments 3 and 4, which took place after the final version of RAP M3 Section 1.2.4/1.25: modify the length/description of optional extension period Section 1.3: modify table 1-1. Section 2.1: clarify the data cutoff; update the SAS version used for analysis Section 2.1.1.7: clarify how proportion will be calculated Section 2.1.1.12: provide criteria of adequate follow-up for time point in the extension period, Section 2.4.1 : remove language 'For first 2 periods, analysis results will be presented in one table; while analysis results from overall study period will be presented in a separate table' Section 2.4.1: change 'dose level' to 'dose' and clarify it is total daily dose. Section 2.7.1: clarify it is 'stratified' log-rank test. Section 2.7.2: clarify that analysis will also include scheduled visits during the extension period with adequate follow-up and last available assessment of the study Section 2.7.3: clarify that for extension period, only scheduled visits with adequate follow-up will be included in the analysis Section 2.7.4: clarify that analysis will also include scheduled visits during the extension period with adequate follow-up Section 2.7.5: clarify that analysis will also include Week 72 and EOT extension Section 2.7.6: clarify that analysis will also include last available assessment. Section 2.8: clarify that the overall study period refers

Date	Time point	Reason for update	Outcome for update	Section and title impacted (Current)
				to overall study period while on LCI699 (e.g. excluding period on placebo) ; remove language 'For first 2 periods, analysis results will be presented in one table; while analysis results from overall study period will be presented in a separate table' Clarifying the calculation of proportion for responder for timepoints in extension (Section 2.1.1.7) Providing criteria of adequate follow-up for timepoints in the extension period (Section 2.1.1.11) Added subgroup analysis for the study (Section 2.2.1) Changes related to PRO section: Section 1.3 table 1.1: clarify that for CushingQoL, in addition to Total score, 2 subscale will also be analyzed Section 2.2: clarify the analysis set used for PRO analysis Section 2.11: provide more details regarding the PRO questionnaires collected in the study. Section 5: new references for PRO were added Changes made following team's review of TFL shell: Section 2.7.4: analysis with adjustment for increased usage in corrective medication were removed. Section 2.8.1.4: update the list of AE summary tables. To correct or improve clarity/consistence of the document, following edits were made: Update the list of abbreviation Core period/core phase was used interchangeable before. Now 'core phase' was used consistently throughout this amendment. Section 2.1.1.11: 'PPS' was corrected to 'PPFAS' Section 2.2: 'treatment' was changed to 'study treatment' Section 2.3.1: for patient disposition, added missed interval between Week 32 and Week 34. Section 2.4.2: Add some standard language regarding definition/coding of concomitant therapy; Section 2.4.2: regarding how to impute the start date of concomitant medication, the language was made more general to cover both 'missing' and 'partially missing'. Section 2.9: the window for vomiting after drug administration was changed from 2hr to 4hr, and some language changes (proposed by CP/CPStat).
22-Apr-2018	Prior to DB lock	Creation of SAP Amendment 2	Clarify that it is the mUFC reported from central lab will be used in all analyses related to mUFC Remove the section 2.1.1.10 (increase use of corrective medication)	Section 2.1.1.4 Section 2.1.1.10

Date	Time point	Reason for update	Outcome for update	Section and title impacted (Current)
06-Jul-2018	After DB lock	Addendum	PDs leading to exclusion of PPFAS/PPRAS	Section 2.2 table 2.3
			Adding summary table for AESI leading to DC and AESI requiring dose adjustment or interruption.	Section 2.8.1.4
			Clarify on tumor volume related analysis	Section 2.8.4.3
			Clarify on acceptable window for Ctrough (provided from PMX group)	Section 2.9
			Change the analysis of 'time-to-last control of mUFC during the randomized withdrawal period' to 'time-to loss of control during the randomized withdrawal period'	Section 4 – Changes to protocol specified analysis

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List of abbreviations

ACTH	Adrenocorticotrophic Hormone
AE	Adverse Event
AESI	Adverse event of special interest
ALP	Alkaline Phosphatase
ALT	Alanine aminotransferase/glutamic pyruvic transaminase/GPT
AST	Aspartate aminotransferase/glutamic oxaloacetic transaminase/GOT
ATC	Anatomical Therapeutic Chemical
BDI-II	Beck Depression Inventory-II
BMI	Body Mass Index
CI	Confidence Interval
CRF	Case Report/Record Form
CRO	Contract Research Organization
CSR	Clinical Study Report
CTC	Common Terminology Criteria
DBP	Diastolic Blood Pressure
DHEAS	Dehydroepiandrosterone
DRL	Drug Reference Listing
DXA	Dual-energy X-ray absorptiometry
ECG	Electrocardiograms
EOS	End of Study
EOT	End of Treatment
FAS	Full Analysis Set
FPG	Fasting Plasma Glucose
HbA1C	Glycosylated Hemoglobin
HPA (-Axis)	Hypothalamic Pituitary Adrenal
IVRS	Interactive Voice Response System
LLOQ	The limit of quantitation
MedDRA	Medical Dictionary for Drug Regulatory Affairs
MID	Minimal Important Difference
MoA	Mechanism of Action
MRI	Magnetic Resonance Imaging
PAS	Pharmacokinetics analysis set
PD	Pharmacodynamics
PRO	Protocol deviations
PoC	Proof of Concept
PPFAS	Per-Protocol Set for Full Analysis Set
PPRAS	Per-Protocol Set for Randomized Analysis Set
PRO	Patient-reported Outcomes
QoL	Quality of Life
QTcF	QT corrected (Fridericia QT formula)
RAS	Randomized Analysis Set
RMP	Risk Management Plans

SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SAS	Safety Analysis Set
SBP	Systolic Blood Pressure
SCS	Summary of Clinical Safety
SOC	System Organ Class
TBIL	Total bilirubin
UFC	Urinary Free Cortisol
ULN	Upper limit of normal range
WHO	World Health Organization

1 Introduction

The SAP document describes the planned statistical analyses for study LCI699C2301 in the treatment of Cushing's disease. It incorporates

- The core phase analysis, which is to be performed after all patients have completed the core phase of the study (Week 48) or discontinued earlier.
- The extension period analysis, which is to be performed after all patients have completed the extension period (Week 72) or discontinued earlier.

1.1 Protocol amendments

At the time of finalization of this document the original study protocol had undergone four amendments, as summarized below.

Amendment 1, 15-JUL-2014

The purpose of this protocol amendment is to address requests from the Voluntary Harmonization Procedure (VHP) review. Changes made to the protocol during this amendment were summarized below:

- The definition of the optional extension period was revised to specify the length of study treatment.
- The definition of time-to-escape (one of the secondary objectives) was clarified.
- To ensure that pregnancy was identified as an absolute withdrawal criterion, it was moved from the list of general study withdrawal criteria to the list of study specific criteria that require study treatment discontinuation.
- To harmonize with ICH guidelines, treatment discontinuation criteria was revised to include an increase in QTcF > 60 msec from pre-first dose baseline measurement.
- To improve monitoring for potential QT prolongation, 24-hour Holter recordings were added during the extension at Weeks 72 and 96. As a result of this addition, both the Visit Evaluation Schedule and the ECG Collection Table were also revised.

Amendment 2, 11-Mar-2015

The primary reason for this amendment is to add a local, country-specific intensive PK sampling for China in order to investigate potential ethnic differences in LCI699 pharmacokinetics at steady-state and at doses used in the treatment of patients with Cushing's disease.

Additional changes applicable to all countries include:

- Inclusion of recent LCI699 clinical trial results information (longer-term safety and efficacy data (LCI699C2201) and results of the clinical drug-drug interaction study (LCI699C2102)).
- Relaxation of the protocol guidance on narrow therapeutic index/sensitive substrates of CYP1A2, CYP2C19, CYP2D6 and CYP3A4/5 as concomitant medication (based on the results of the clinical drug-drug interaction study (LCI699C2102)).

- Blinding: The pharmacist, the bioanalyst and the pharmacokineticist are also blinded.
- Inclusion criteria:
 - Minimum period since last stereotactic radiosurgery decreased from 3 to 2 years.
 - Rescreening is introduced, some therapies needed washouts are removed and the screening period is extended to accommodate for the need for long washout periods.
- Exclusion criteria:
 - QTcF exclusion limits tightened from >470 ms to >450 ms (males) and >460 ms (females).
 - Definition of post-menopausal status and woman not of childbearing potential is added.
 - Male contraception is no longer required in patients treated with LCI699 based on a thorough review of the mechanism of action of LCI699 and re-evaluation of pre-clinical and clinical data.
 - The criterion on optic chiasm compression is updated and broaden.
 - Euthyroid status is based on investigator's judgment (rather than biochemically).

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- Changes to safety monitoring schedule (removal of non-required assessments timepoints, additional sex hormone assessments, assessment of waist circumference)
- Collection of additional baseline information (such as information on most recent prior medical therapy).
- Permitted concomitant medications now includes spironolactone, eplerenone, cyproterone acetate or finasteride under certain conditions.
- Discontinuation criteria revised to distinguish “confirmed” laboratory abnormalities and “confirmed and persistent laboratory abnormalities”; with the hypokalemia parameter relaxed.
- Central reading of photographs removed, as it is not essential for the study.
- Additional sex hormone assessments (Androstenedione, DHEA and Estrone) are introduced to better characterize the effects on hormone balance.
- The CTCAE version is updated from 4.0 to 4.03.
- The “potential risk” for QT prolongation is changed to “risk”, based on one reported SAE with QTcF prolongation in a clinical study evaluating the longer-term efficacy and safety of LCI699 (LCI699C2201).
- The QT monitoring section is clarified and the trigger for doing a triplicate ECG is updated from > 500 msec to > 480 msec.

- Novartis Discontinuation of Clinical Trial Protocol Elements language has been implemented throughout the protocol.

In addition, editorial changes and clarifications were made throughout the protocol.

As of 02 March 2015, 16 patients have provided written informed consent for the study and 10 patients have received LCI699.

Amendment 3, 29-Mar-2016

The purpose of this amendment is to include the following changes to reduce the risk of dosing errors.

- Expand the description of the dose dispensation process; in particular to emphasize that more than one tablet strength of study drug may be dispensed at the same visit.
- Elaborate on dose adjustments and communication of dosing instructions
- Recommend the use of patient dose instruction card and phone contacts between scheduled visits.

This amendment also includes changes to ensure patient safety by adding specific criteria for the identification and management of patients with potential drug-induced liver injury (DILI). Although there are no known cases of suspected DILI in patients treated with LCI699 to date, these criteria are added in the event that a case of suspected DILI arises in the future.

Other protocol changes are:

- To provide more information regarding the study drug:
 - A summary of the benefit-risk assessment of LCI699 for this trial has been added
 - The results of the thorough QT/QTc study (LCI699C2105) of LCI699 in healthy volunteers has been added
- Additional measurements of serum cortisol and plasma ACTH have been added for patients in China undergoing extensive PK, [REDACTED]
- Exclusion criterion # 16 is modified to exclude patients with serum total bilirubin $> 1.5 \times$ ULN; this is one of the criteria needed for effective screening of patients for potential DILI

The duration of the optional extension period is increased in order to collect additional long-term safety data as well as to provide continued access to the study drug for those patients benefitting from the treatment. Patients can continue on extension period until the last patient completes the core phase or discontinued early from core phase. Patients who continue to benefit from treatment will be offered participation in a separate long-term safety follow-up study. Study CLCI699C2301 ends when seamless transition to the long-term safety follow-up study is possible or alternative treatment options are available; this period will not exceed 4 months.

Amendment 4, 06Jul-2017

The main purpose of this amendment is to further increase the duration of the optional extension period in order to collect additional long-term safety and efficacy data as well as to provide continued access to the study drug for those patients benefitting from the treatment until a separate long-term safety follow-up study is set up at participating sites. Based on this extension, the end of study definition has been updated. In addition, the long-term safety follow-up study modalities have been detailed.

As a result of the change in the duration of the optional extension period, the study objectives and associated statistical sections have been revised to include extension period time points.

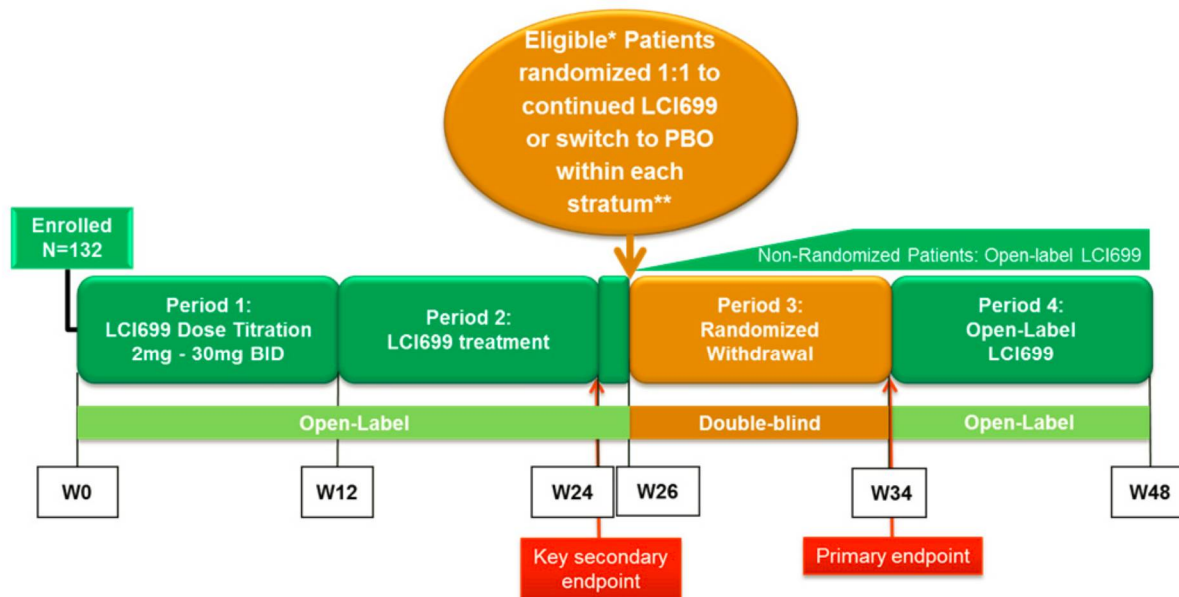
Other protocol changes include:

- In view of the results of the thorough QT study CLCI699C2105, which showed that the increase in QTcF caused by LCI699 at therapeutic doses is below the threshold of regulatory concern, the QT-specific concomitant medication guidance for LCI699 was revised to limit the list of prohibited drugs to medications with a “Known risk to cause TdP” and “Possible risk to cause TdP”, instead of all drugs known to prolong QT. This change is also in alignment with the terminology used in the QT Drug Lists (CredibleMeds®). The risks section was updated to include neutropenia, which is a known effect related to the decrease of cortisol in patients with Cushing’s disease, in line with cases observed in clinical trials with LCI699.

1.2 Study design

This is a Phase III, multi-center, double-blind, randomized withdrawal study following a 24 week, single-arm, open-label dose titration and treatment period to evaluate the safety and efficacy of LCI699 for the treatment of patients with Cushing’s disease. The study has a core phase (consists of four study periods 1-4, see schematic diagram below) and an optional extension period, which are described below.

Figure 1-1 Schematic of core study design

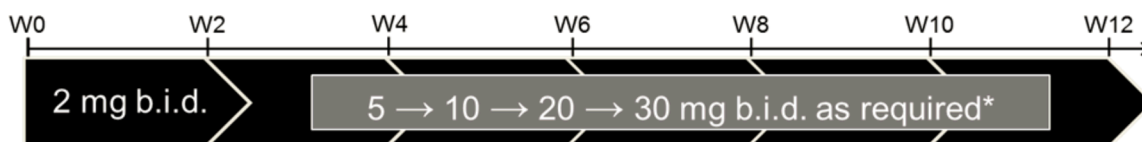


*To be eligible for randomization, the patient must have mUFC $\leq$ ULN at Week 24, and no further dose increase after Week 12.
**Strata are determined by the combination of two stratification factors at randomization: 1) LCI699 dose at Week 24 ($\leq$ 5mg bid vs. > 5mg bid), and 2) history of pituitary irradiation (yes/no).

1.2.1 Study Period 1

This single-arm, open-label LCI699 period (Week 1 to Week 12) is the individual patient dose titration period.

Figure 1-2 Schematic of study period 1



* Dose can be down titrated to 1 mg b.i.d. if needed

Dose adjustments are based on the mean of three 24-hour UFC (mUFC) values as measured by the central lab. The triplicate urine samples are collected every two weeks during individual dose titration with the last urine sample preferably collected the day prior to the visit at site. The dose is increased if mUFC is above normal ($>$ ULN). The dose is reduced if mUFC $<$ LLN, or if the patient is symptomatic and mUFC is in the lower part of the normal range. The dose should be maintained if mUFC is within the normal range and the patient does not have signs or symptoms of adrenal insufficiency. At Week 0 and Week 2, dose increase is not permitted.

1.2.2 Study Period 2

This period (Week 13 to Week 26) aims to assess the efficacy and safety of LCI699 at the therapeutic dose determined during the dose titration period. Patients whose mUFC becomes elevated during this period can have their LCI699 dose increased further, if it is tolerated and the maximum dose of 30 mg b.i.d. has not been reached. Such patients will be followed for long-term safety but will not be considered complete responders, and cannot be randomized. Dose reductions and temporary dose interruptions for safety reasons are permitted during this period without affecting eligibility for randomization.

In order for a patient to be classified as a complete responder at Week 24, the following two conditions must be met: 1) mUFC $\leq$ ULN based on urine samples collected at Week 24; 2) LCI699 dose during Study Period 2 (Weeks 13-24) was not increased above the level established at the end of Study Period 1 (Week 12; end of individual dose titration period). Dose reductions and temporary dose interruptions for reasons of safety do not preclude the possibility of complete response assessment at Week 24.

The key secondary endpoint, the proportion of patients with a complete response (mUFC $\leq$ ULN), is assessed at Week 24.

Patients remain on open-label LCI699 during the period between week 24 and week 26, in order to ensure that sufficient time is allowed for central laboratory results (Week 24 mUFC) to become available for all patients at all sites, and to standardize the time of randomization across sites.

1.2.3 Study Period 3

Study Period 3 is a double-blind, placebo-controlled randomized withdrawal period (Week 26 to Week 34).

Double blinding

The Novartis study team, the patient, the investigator, and all other site staff remain blinded to treatment assignment from the time of randomization to the time of database lock for the core phase of the study.

Eligibility for randomization

In order to be eligible for randomization, patients must have completed dose titration during the first 12 weeks, continued on LCI699 treatment with no further dose increase during Weeks 13-24, and have normal UFC (mUFC $\leq$ ULN) from urine samples collected at Week 24. Randomization is implemented at the Week 26 visit.

Patients that are not eligible for randomization are followed on open-label LCI699 until the end of the core treatment (Week 48), unless there is a reason to discontinue from the study prematurely.

Randomization

Eligible patients are randomized in a double-blinded fashion at Week 26 at a 1:1 ratio either to continue treatment with LCI699 at the same dose or to matching placebo. Patients are stratified at randomization according to: LCI699 dose at Week 24 (≤ 5 mg b.i.d. vs. > 5 mg b.i.d.), and history of pituitary irradiation (yes/no).

UFC monitoring during randomized withdrawal

During the 8-week randomized withdrawal study period, mUFC is measured at scheduled visits every 2 weeks. However, patients are also allowed to have unscheduled visits at any time during the randomized withdrawal period if they report symptoms of hypercortisolism or hypocortisolism. The investigator decides the dose of study drug (LCI699 or placebo) during this period, although is blinded to treatment assignment. All laboratory tests during the randomized withdrawal period must be sent to the central laboratory for analysis, and all treatment decisions must be based on central laboratory results.

Dose adjustments during randomized withdrawal

The dose of study drug (LCI699 or placebo) should remain unchanged for patients that maintain a normal mUFC and do not develop AE's related to study drug during randomized withdrawal. The investigator may reduce or withhold a dose of study drug for safety reasons at any time during the study, including the randomized withdrawal period. Dose reductions or interruptions for safety reasons during the randomized withdrawal period do not preclude the possibility of a complete response at Week 34. Dose increases are not permitted during the randomized withdrawal period.

Discontinuation from randomized withdrawal

During the randomized withdrawal study period, the patient must be discontinued from the randomized withdrawal period, declared a non-responder, if the mUFC increases to $> 1.5 \times$ ULN, and at least 2 individual urine samples show UFC $> 1.5 \times$ ULN at a single visit (scheduled or unscheduled).

After discontinuation from randomized withdrawal, or at the end of the randomized withdrawal period (Week 34), whichever comes first, the patient resumes open-label LCI699 at a dose selected by the investigator.

Patients who are discontinued from the study during the randomized withdrawal period are no longer in the study, and consequently they are not permitted to receive open-label LCI699 and cannot move forward to Study Period 4.

1.2.4 Study Period 4

Study Period 4 is a single-arm, open-label therapy (Week 35 to Week 48). After Week 34 visit, all patients receive open-label LCI699 treatment until Week 48. Dose changes are permitted.

At Week 48, patients have the option to enter an extension period or discontinue LCI699 at week 48 to conclude with an end of core study visit 4 weeks off study drug (at Week 52).

1.2.5 Optional extension period

Patients who continue to receive clinical benefit, as assessed by the study investigator and who wish to enter the extension period must be re-consented at week 48. Patients who enter the extension period will do so without interruption of study drug or assessments. The optional extension period will end after all patients have completed Week 72 or have discontinued early (prior to Week 72).

Study CLCI699C2301 ends when all ongoing patients have transitioned to the long-term safety follow-up study or have been offered local alternative treatment options; this period will not exceed 4 months after all ongoing patients have completed Week 72. Patients who continue to benefit from treatment and have completed Week 72 will be offered participation in a separate long-term safety follow-up study after the database lock of the core phase is completed.

If the long-term safety follow-up study is opened at site, patients should transition to the long-term safety follow-up study. If this option is not available, the patient can either be offered a local alternative treatment option or stay in the study until the study end (i.e. 4 months after the last patient completes Week 72). Patients entering the long-term safety follow-up study will complete an EOT visit. For these patients, the EOS visit is not applicable, as treatment on LCI699 will not be interrupted. Patients not entering the long-term safety follow-up study will complete an EOT visit, and an EOS visit 30 days after the last dose administration.

1.3 Study objectives and endpoints

The following table displays the study objectives/endpoints and the sections of this SAP document that describe the corresponding analyses.

Table 1-1 Study Objectives and endpoints

Objective	Endpoint	Analysis Section
Primary		
To compare the complete response rate at the end of the 8-week period of randomized withdrawal (Week 34) between patients randomized to continued LCI699 therapy vs. placebo.	Proportion of randomized patients in each arm with: $mUFC \leq ULN$ at the end of 8 weeks of randomized withdrawal (Week 34), and were neither discontinued, nor had LCI699 dose increase above the level at week 26 during the randomized withdrawal period.	2.5.1
Key secondary		
To assess the complete response rate at the end of individual dose-titration and treatment with LCI699 in the initial single-arm, open label period (Week 24).	Proportion of enrolled patients with $mUFC \leq ULN$ at Week 24 and had no dose increase above the level established at Week 12 between Week 13 and Week 24.	2.6.1

Objective	Endpoint	Analysis Section
Other secondary		
To compare the time-to-last control of mUFC during the randomized withdrawal period between patients randomized to continued LCI699 therapy and placebo.	Time-to-last control of mUFC, which is defined as the time (in days) from randomization to the last mUFC collection that was $\leq$ ULN before early discontinuation or completion of randomized withdrawal period, whichever is earlier.	2.7.1
To assess the complete, partial, and overall response rate at Week 12, Week 24, Week 48, and at scheduled time points during the extension period and the last available assessment.	Complete response rate: proportion of enrolled patients with mUFC $\leq$ ULN at Week 12, Week 24, Week 48, and at scheduled time points during the extension period (provided adequate follow-up), and the last available assessment Partial response rate: proportion of enrolled patients with $\geq 50\%$ reduction from baseline in mUFC, but mUFC $>$ ULN) at Week 12, Week 24, and Week 48, and at scheduled time points during the extension period (provided adequate follow-up), and the last available assessment Overall response rate: proportion of enrolled patients with mUFC $\leq$ ULN or at least 50% reduction from baseline at Week 12, Week 24, Week 48, and at scheduled time points during the extension period (provided adequate follow-up), and the last available assessment	2.7.2
To assess the change in mUFC during the core and extension periods of the study.	- Actual and percentage change in mUFC from baseline to each post-baseline visit during the core and extension (provided adequate follow-up) at which UFC is collected - Actual and percentage change in mUFC from the time of randomization (Week 26) to the end of the randomized withdrawal period (Week 34), or the last mUFC measurement prior to early discontinuation, whichever occurs earlier.	2.7.3
To assess the change in cardiovascular-related parameters associated with Cushing's disease during the core and extension periods of the study.	- Actual and percentage change from baseline during the core and extension periods (provided adequate follow-up) of the study in: fasting glucose, HbA1c, fasting lipid profile, blood pressure, body weight, BMI and waist circumference - Actual and percentage change from the randomization (Week 26) to the end of randomized withdrawal period (Week 34), or the last measurement available prior to early discontinuation, whichever occurs earlier (see bullet above for individual parameters).	2.7.4
To assess the change from baseline in the physical features of Cushing's disease by photography at Week 12, 24, 34, 48, and during the extension period.	Categorical change from baseline to Week 12, 24, 34, 48, during the extension at week 72 and EOT extension in each of the following clinical signs of Cushing's disease by photography: facial rubor, hirsutism, striae, supraclavicular fat pad, dorsal fat pad, proximal muscle wasting (atrophy), central	2.7.5

Objective	Endpoint	Analysis Section
	(abdominal) obesity, and ecchymoses (bruises).	
To assess the change from baseline in bone mineral density by DXA scan at the lumbar spine and total hip at Week 48 and the last available assessment.	Actual and percent change from baseline to Week 48 and the last available assessment in bone mineral density as measured by DXA scan at the lumbar spine and total hip	2.7.6
To assess the time-to-escape.	Time-to-escape is defined as the time (in days) from the first mUFC $\leq$ ULN to the first mUFC results $> 1.5 \times$ ULN with at least 2 individual UFC results $> 1.5 \times$ ULN.	2.7.7
To assess general safety and AEs of special interest	Adverse events and laboratory abnormalities will be assessed using the National Cancer Institute-Common Toxicology Criteria (NCI-CTC) grading scale (version 4.03). AEs of special interest, as reported by the investigator, or by laboratory evaluation, ECG, Holter recording, and pituitary MRI.	2.8
To evaluate exposures of LCI699 in patients with Cushing's disease	Plasma concentrations (predose, 0.5 h, 1.5 h, and 3.5 h post-dose) of LCI699	2.9.1
To assess the change in Patient-Reported Outcomes (Health-Related Quality of Life) during the core and extension periods of the study.	- Change in standardized score of CushingQoL (to include 2 subscale and total scores), Beck Depression Inventory-II, and EQ-5D-5L (utility index and VAS), from baseline to Week 24 and Week 48. - Change in standardized score of CushingQoL (to include 2 subscale and total scores), Beck Depression Inventory-II, and EQ-5D-5L (utility index and VAS), from the randomization (Week 26) to the end of randomized withdrawal period (Week 34), or the last measurement prior to early discontinuation, whichever occurs earlier. - Change from baseline in standardized score of CushingQoL (to include 2 subscale and total scores), Beck Depression Inventory-II, and EQ-5D-5L (utility index and VAS), from baseline to Week 72, 96 and the EOT extension	2.11.1

2 Statistical methods

2.1 Data analysis general information

Patients who discontinue study drug will be considered withdrawn from the study after the final visit assessments and the final safety follow-up visit are performed or when it is clear that the patient will not return for final visit assessments and/or final safety follow-up visit.

The primary analysis will be performed based on cumulative data collected up to the date when all enrolled patients have either completed or prematurely withdrawn from the core phase of study. The results and outcomes of this analysis will be presented in an interim CSR. All additional data collected subsequently to the cutoff date for the primary analysis will be included in the analysis for the final CSR, which will be performed once all patients have either completed the extension period or discontinued earlier.

Novartis or designated CRO will analyze all data using the SAS System for data analysis V9.4 or higher. The data from all centers participating in the trial will be combined, so that an adequate number of patients will be available for analysis.

2.1.1 General definitions

2.1.1.1 Study day

Study Day 1 is the date of first administration of study drug.

The study day for an event that occurred prior to study day 1 will be calculated as (date of event – date of first administration of study drug).

The study day for an event that occurred after study day 1 will be calculated as (date of event – date of first administration of study drug) +1.

Study day will be displayed in the data listings.

2.1.1.2 Baseline

For efficacy and safety evaluations (e.g. laboratory assessments and vital signs): the last available pre-dose assessment within 35 days prior to or on Study Day 1 (before the first dose of LCI699) is taken as the “baseline” assessment; the last available assessment within 21 days prior to the first dose of randomized treatment (LCI699 or Placebo) is taken as the “randomization” assessment; If patients have no value as defined above, the baseline/randomization result will be considered missing.

2.1.1.3 Visit number

The time point (Day/Month) associated with an assessment will be determined by the visit number assigned to the assessment in the database. The mapping of visit numbers to time points in the core study and the extension study are provided in the table below.

Table 2-1 Scheduled visits in core phase

Core study Visit	Day/Week	Clarifying Notes
Visit 1	Day -35 to Day -8	Screening
Visit 2	Day -7 to Day -1	Baseline
Visit 3	Day 1/Week 0	Dose escalation period
Visit 4	Day 15/Week 2	Dose escalation period
Visit 5	Day 29/Week 4	Dose escalation period
Visit 6	Day 43/Week 6	Dose escalation period
Visit 7	Day 57/Week 8	Dose escalation period
Visit 8	Day 71/Week 10	Dose escalation period
Visit 9	Day 85/Week 12	Dose escalation period
Visit 10	Day 113/Week 16	Treatment period
Visit 11	Day 141/Week 20	Treatment period
Visit 12	Day 169/Week 24	Treatment period
Visit 13	Day 183/Week 26	Randomized Withdrawal period
Visit 14	Day 197/Week 28	Randomized withdrawal period
Visit 15	Day 211/Week 30	Randomized withdrawal period
Visit 16	Day 225/Week 32	Randomized withdrawal period
Visit 17	Day 239/Week 34	Randomized withdrawal period
Visit 18	Day 253/Week 36	Open Label period
Visit 19	Day 281/Week 40	Open Label period
Visit 20	Day 309/Week 44	Open Label period

Table 2-2 Scheduled visits in extension period

Extension study Visit	Day/Week	Clarifying Notes
Visit 21	Day 337/Week 48	Extension
Visit 22	Day 365/Week 52	Extension
Visit 23	Day 393/Week 56	Extension
Visit 24	Day 421/Week 60	Extension
Visit 25	Day 449/Week 64	Extension
Visit 26	Day 477/Week 68	Extension
Visit 27	Day 505/Week 72	Extension
Visit 28	Day 589/Week 84	Extension
Visit 29	Day 673/Week 96	Extension

Extension study Visit	Day/Week	Clarifying Notes
Visit 21	Day 337/Week 48	Extension
Visit 30	Day 757/Week 108	Extension
Visit 31	Day 841/Week 120	Extension
.....	Every 84 days/12 weeks	Extension

Unlike the above visits that are scheduled to occur after a fixed number of days have elapsed since Day 1, the Visits 777 (End-of-Treatment in core phase) and 778 (End-of-Treatment in extension period) are scheduled when the patient either completes the respective study period or decides (or is mandated by the protocol) to prematurely discontinue from the study.

If such a visit happens neither too soon nor too late after the actual last scheduled visit (LASTVIS) prior to the 777/778 visit, then it will be mapped to the next scheduled visit (LASTVIS+1) in the visit schedule (from the tables above).

The 777/778 visit will be considered to have occurred neither too soon nor too late after the actual last scheduled visit if the number of days between the two visits is

(i) at least half of the planned gap between the patient's actual last scheduled visit (LASTVIS) and the next scheduled visit and

(ii) at most the total of (a) the planned gap between the patient's actual last scheduled visit (LASTVIS) and the next scheduled visit and (b) half of the planned gap between the patient's next two scheduled visits.

If the number of days between the 777/778 visit and LASTVIS is less than the range specified above then the 777/778 visit will be mapped to an unscheduled visit of LASTVIS; if it is more then it will be mapped to an unscheduled visit of LASTVIS + 1.

2.1.1.4 Calculating mean UFC

To compute the mean UFC for a patient at any particular visit, at least two individual 24hr UFC specimens are required at that visit. If there are less than two samples available then mean UFC will be considered missing for that assessment. The mUFC value calculated by central lab following this rule will be used in all analyses related to mUFC (Protocol section 10.4.3).

2.1.1.5 Study day associated with a UFC assessment

The study day associated with a UFC assessment at any particular visit is defined as the study day of the last UFC sample collection for that visit.

2.1.1.6 Conversion of duration in days to duration in months/years

Duration in months = $12 * (\text{Duration in days}) / 365.25$

Duration in years = $(\text{Duration in days}) / 365.25$

2.1.1.7 Calculation of proportions of responders

All enrolled patients will have either reached Week 48 or discontinued earlier at the cutoff date for the database lock for the core phase analysis. Therefore, all enrolled patients will be included in the analysis for calculating the proportion of responders at time points up to Week 48.

For calculation of proportion of responders beyond Week 48 (e.g. a time point in extension period), the following rules will be applied to determine whether patients are included in the analysis.

- 1) patients who declined to enter optional extension period after completion of the core phase will always be excluded.
- 2) patients who are still ongoing at the cutoff date for the data base lock will be include for time points up to their last available scheduled visit. For example, an ongoing patient with last scheduled visit Week 52 at the cutoff date will be included in the analysis for time points up to Week 52.
- 3) patients who discontinued prior to the data cutoff date will be included up to the furthest scheduled visit they could have completed if they did not discontinued early based on data cut off and last completed based on the date of last completed schedule visit and analyses cut-off date. For example, if the last completed scheduled visit for an early discontinued patient is Week 52, and the time between data cutoff and last completed scheduled visit is 8 weeks, he/she will be included in the analyses up to Week 60.

If included in the analysis for calculating the proportion of responders at a given timepoint, patients discontinued before this timepoint will be counted as non-responders.

2.1.1.8 Method for calculating Confidence Intervals

95% confidence intervals for proportions will be calculated using the exact (Clopper-Pearson) method, unless stated otherwise.

95% confidence intervals for change and percentage change from baseline will assume normally distributed data and will be calculated using the t-distribution.

2.1.1.9 AE summary cutoff

All data up to data cut-off date will be at least listed. All safety data prior to the data cut-off will be summarized with the exception of AEs and deaths, which will follow the cut-off rule specified below.

- AE/death summary cut-off date = min { data cut-off date, max(last dosing date + 30 , safety follow-up visit date) }

AEs/deaths that occur after the AE/death summary cut-off date but prior to the data cut-off date will be listed and flagged.

2.1.1.10 Criteria of adequate follow-up for time points in the extension period

For a given time point in the extension period, it will be considered to have adequate follow-up to be included in the reporting activities if there are at least 15 patients available for the analysis at that time point.

2.2 Analysis sets

Randomized analysis set (RAS): comprises all randomized patients who have received at least one dose of randomized drug (LCI699 or placebo). Following the ITT principle, patients will be analyzed according to the treatment and stratum they have been assigned to during the randomization. RAS is used in efficacy analysis for RW period only including the primary endpoint.

Full analysis set (FAS): comprises all enrolled patients who receive at least one dose of LCI699. FAS is used in efficacy analysis for study period other than RW period including key secondary endpoint.

Safety set: There are two safety sets defined in this study. Safety analysis set (SAS) comprises all enrolled patients who received at least one dose of LCI699 and had at least one valid post-baseline safety assessment. Safety Analysis Set for randomized withdrawal period (SASR) comprises only randomized patients who received at least one dose of randomized treatment (LCI699 or placebo) and had at least one valid safety assessment during the randomized withdrawal period. Please note: the statement that a patient had no adverse events (on the Adverse Event CRF) constitutes a safety assessment. SASR is used for safety analysis for RW period only, while SAS is used in safety analysis for study period other than RW period only. For safety analysis, patients will be analyzed according to the study treatment received, where study treatment received is defined as the randomized/assigned treatment if the patient took at least one dose of that treatment or the first treatment (first dose of the study for SAS and first dose of randomized withdrawal period for SASR) received if the randomized/assigned treatment was never received.

Per-Protocol set: There are two per-protocol sets defined in this study. Per-Protocol Set for Randomized Analysis Set (PPRAS) consists of a subset of the patients in the RAS who had no selected protocol deviation. Per-Protocol Set for Full Analysis Set (PPFAS) consists of a subset of the patients in the FAS who had no selected protocol deviation. PPFAS will only be used in supportive analyses for key secondary efficacy endpoint, while PPRAS will only be used in supportive analyses for primary efficacy endpoint. Details of protocol deviations that lead to exclusion from PPS and/or PPRAS can be found in table below.

Table 2-3 Protocol deviations leading to exclusion from PPFAS and/or PPRAS

Protocol Deviation description	Protocol deviation Code(s) in VAP3 (Amendment 4, July2017)	Excluded from PPFAS?	Excluded from PPRAS?
Informed consent for core phase not documented	I01	Yes	Yes
mUFC related inclusion criteria not met	I06/I07	Yes	Yes

ACTH related inclusion criteria not met	I09	Yes	Yes
Cushing's disease confirmation is missing	I10	Yes	Yes
Pituitary surgery related inclusion criteria not met	I11/I12	Yes	Yes
Pituitary irradiation related inclusion criteria not met	I14	Yes	Yes
Washout period related inclusion criteria not met	I13/I15/I16/I17/I18/I19/I20	Yes	Yes
Subject did not meet the eligibility criteria for randomization but was randomized at Week 26.	I21	No	Yes
Participation in another trial with an investigational drug does meet the exclusion criteria	E01	Yes	Yes
Subject pregnant or lactating at study entry	E04/E05	Yes	Yes
Subject's history of hormone over secretion does meet the exclusion criteria	E10	Yes	Yes
Subject's medical history of Cushing's syndrome does meet the exclusion criteria	E11	Yes	Yes
Subject compliance history does meet the exclusion criteria	E23	Yes	Yes
Subject met the discontinuation criteria, but was not discontinued from study treatment	D01/D02	Yes*	Yes**
Subject took a prohibited concomitant medication	M01	Yes*	Yes**

*If protocol deviation happens before Week 24, when the key secondary endpoint is assessed

**If protocol deviation happens before end of randomized withdrawal period, when the primary endpoint is assessed

Pharmacokinetic analysis set (PAS): consists of all enrolled patients who receive at least one dose of LCI699 and have at least one evaluable post dosing PK assessment. A PK concentration or parameter can be considered non-evaluable as per scientific judgment of the clinical pharmacology expert; reasons will be documented (see Section [2.9](#))

The analysis sets used for various analyses are summarized in the following table:

	FAS	RAS	SAS/SASR	PPFAS/ PPRAS	PAS
Patient Disposition	√	√			
Demography and baseline disease characteristics	√	√			
Medical History	√				
Prior Medication	√				
Protocol Deviation	√	√			

Concomitant Medications			√	
Exposure to study medication			√	
Efficacy analyses	√	√		√*
Safety analyses			√	
Patient-reported outcomes	√	√		
PK analysis				√

* PPFAS will only be used in supportive analyses for key secondary efficacy endpoint, while PPRAS will only be used in supportive analyses for primary efficacy endpoint.

When appropriate, patients in a given analysis set could be classified into mutually exclusive treatment groups as described below:

For patients in RAS/PPRAS

- LCI699 group: patients randomized to continued treatment of LCI699 for the randomized withdrawal period of the study
- Placebo group: patients randomized to placebo for the randomized withdrawal period of the study

For patients in FAS/PPFAS

- LCI699 group: patients randomized to continued treatment of LCI699 for the randomized withdrawal period of the study
- Placebo group: patients randomized to placebo for the randomized withdrawal period of the study
- Non-randomized patients: patients who do not enter randomized withdrawal period of the study (either discontinued study before the randomization or not eligible to be randomized)

For patients in SAS

- LCI699 group: patients receiving study treatment of LCI699 for the randomized withdrawal period of the study
- Placebo group: patients receiving study treatment of placebo for the randomized withdrawal period of the study
- Non-randomized patients: patients who do not enter randomized withdrawal period of the study (either discontinued study before the randomization or not eligible to be randomized)

For patients in SASR

- LCI699 group: patients receiving study treatment of LCI699 for the randomized withdrawal period of the study
- Placebo group: patients receiving study treatment of placebo for the randomized withdrawal period of the study

2.2.1 Subgroup of interest

Subgroup of interest for this study includes the two stratification factors for randomization: history of pituitary irradiation, and LCI699 dose at Week24.

History of pituitary irradiation: Yes and No

LCI699 dose at Week24: $\leq 5\text{mg b.i.d.}$ and $> 5\text{mg b.i.d.}$

For each of the above two stratification factors for randomization (strata as per IVRS), primary efficacy endpoint will be analyzed by randomized treatment group.

2.3 Patient disposition, demographics and other baseline characteristics

Summary statistics of data pertaining to subject disposition, demographics, baseline characteristics, medical history, prior medications and protocol deviations will be tabulated and displayed using grouping described in Section [2.2](#).

2.3.1 Patient disposition

For overall study period, counts of patients for the following items will be summarized by treatment group using FAS:

- discontinued at any time*
- Primary reason for discontinuation at any time
- discontinued at or prior to Week 12
- Primary reason for discontinuation at or prior to Week 12
- discontinued at or prior to Week 24 but after Week 12
- Primary reason for discontinuation at or prior to Week 24 but after Week 12
- discontinued prior to Week 48 but after Week 24
- Primary reason for discontinuation prior to Week 48 but after Week 24
- Completed Week 48
 - Completed Week 48 and did not enter Extension period
 - Completed Week 48 and enter Extension period
- Ongoing in Extension
- Discontinuation in the Extension
- Primary reason for discontinuation in the Extension
- Discontinued at or prior to Week 72 but after Week 48
- Discontinued prior to Week 96 but after Week 72
- Discontinued after Week 96

- Completed extension

* Patients who completed Week 48 and declined to enter optional extension period are not counted as discontinuations.

For randomized withdrawal period, counts of patients for the following items will be provided in a separate disposition summaries presented by treatment group using RAS:

- discontinued from randomized treatment before completing RW period
- primary reason for discontinuation before completing RW period
- discontinued from randomized treatment at or prior to Week 28
- primary reason for discontinuation at or prior to Week 28
- discontinued from randomized treatment at or prior to Week 30 but after Week 28
- primary reason for discontinuation at or prior to Week 30 but after Week 28
- discontinued from randomized treatment at or prior to Week 32 but after Week 30
- primary reason for discontinuation at or prior to Week 32 but after Week 30
- discontinued from randomized treatment at or prior to Week 34 but after Week 32
- primary reason for discontinuation at or prior to Week 34 but after Week 32
- Completed the RW period (Week 34)

2.3.2 Demographic and baseline disease characteristics

All demographic and baseline disease characteristics will be summarized using FAS by treatment group and listed in detail. Categorical data will be summarized by frequencies and percentages. For continuous data, mean, standard deviation, minimum, 25th, median, 75th percentiles and maximum will be presented.

Demographic variables include Age (in year), Age group ($< 65/\geq 65$), gender, race, and ethnicity.

Baseline disease characteristics include Time to first LCI699 dose since diagnosis of Cushing's disease, Cushing's disease status (de novo or persistent/recurrent), previous pituitary surgery/irradiation, mean UFC at baseline.

If the diagnosis date of Cushing's disease is missing or partially missing, it will be imputed following Novartis standard rule for table and listing.

2.3.3 Medical history

Relevant medical history and ongoing medical conditions will be summarized and listed. The summary will be presented using FAS by primary system organ class (SOC), preferred term (PT), and treatment group.

2.3.4 Protocol deviations

The number and percentage of patients with any PDs up to data cutoff will be tabulated by treatment group using FAS.

PDs leading to exclusion from the PPFAS and PPRAS will be tabulated by treatment group using FAS and RAS respectively.

All PDs up to data cutoff will be listed.

The criteria for protocol deviation are specified in Validation and Planning (VAP) Module 3 and will be finalized prior to database lock.

2.4 Treatments (study treatment, rescue medication, concomitant therapies, compliance)

2.4.1 Study treatment / compliance

The exposure to the study drug will be summarized based on the study periods: study period Week 26 weeks (SAS by All patients only), randomized withdrawal period (SAS by LCI699 group and Placebo group), and overall study period (SAS by LCI699 group, Placebo group, Non-randomized patients, and All patients).

Duration of exposure

For study period up to Week 26, duration of exposure to LCI699 for all patients in SAS is calculated as:

$\text{Duration of exposure (weeks)} = (\min(\text{date of Week 26 visit, date of last administration of LCI699, date of data cut-off, date of death}) - \text{date of first administration of LCI699} + 1) / 7$

- For Randomized withdrawal period, duration of exposure to randomized treatment (either LCI699 or Placebo) for patients in SASR is calculated as:

$\text{Duration of exposure (weeks)} = (\min(\text{date of last administration of randomized treatment, date of data cut-off, date of death}) - \text{date of first administration of randomized treatment} + 1) / 7$

- For overall study period, duration of exposure to LCI699 for patients in SAS is calculated as below:

- for Placebo group:

$\text{Duration of exposure (weeks)} = (\min(\text{date of last administration of study medication, date of data cut-off, date of death}) - \text{date of first administration of LCI699} + 1) / 7 - \text{Duration of exposure to Placebo}$

- for LCI group and non-randomized patients

$\text{Duration of exposure (weeks)} = (\min(\text{date of last administration of LCI699, date of data cut-off, date of death}) - \text{date of first administration of LCI699} + 1) / 7$

The above duration calculation includes the periods of temporary interruption.

Dose reduction/interruption

The number of patient who has at least one dose reductions or interruptions, and the corresponding reasons will be summarized by study period. Although patients may have multiple dose reduction or interruption with the same reason, patient will be counted only once for each distinct reason.

The actual and planned doses administered and reason for dose change will be listed with UFC information.

Dose of LCI699

For total daily dose of LCI699, the highest dose, average dose, and the dose with longest duration for each patient will be summarized by study period. In addition, the LCI699 average total daily dose by visit during the core study period will be summarized using descriptive statistics and displayed graphically.

2.4.2 Prior, concomitant and post therapies

Concomitant therapy is defined as all interventions (therapeutic treatments and procedures) other than the study treatment administered to a patient coinciding with the study treatment period. Concomitant therapy include medications (other than study drugs) starting on or after the start date of study treatment or medications starting prior to the start date of study treatment and continuing after the start date of study treatment.

Concomitant medications will be coded using the World Health Organization (WHO) Drug Reference Listing (DRL) dictionary that employs the WHO Anatomical Therapeutic Chemical (ATC) classification system. Surgical and medical procedures will be coded using MedDRA and summarized by SOC and preferred term.

Concomitant medications and significant non-drug therapies will be listed and summarized using frequency counts and percentages by ATC class, preferred term and treatment group within each of three study periods: first 26 weeks (FAS), randomized withdrawal period (RAS), and overall study period (FAS).

Therapies for Cushing's Disease taken prior to first dose of LCI699 will be summarized using frequency counts and percentages by ATC class, preferred term and treatment group.

If the start date of a concomitant medication is missing or partially missing, then it will be imputed following Novartis standard rule for table and listing.

2.5 Analysis of the primary objective

The primary objective is to compare the complete response rates at the end of the 8 weeks period of randomized withdrawal (Week 34) between patients randomized to continued LCI699 therapy vs. placebo.

The primary efficacy variable is the proportion of randomized patients in each treatment arm that are complete responders at the end of the 8 weeks of the randomized withdrawal period (Week 34). A complete responder is defined as a patient who has $mUFC \leq ULN$ (based on central laboratory result) at Week 34 and who was neither discontinued nor had LCI699 dose increase above the level at week 26 during the randomized withdrawal period of the study. Patients who discontinued during the randomized withdrawal period will be counted as non-responders for the primary endpoint. Dose reductions and temporary dose interruptions for safety reason during randomized withdrawal period do not preclude patients from being complete responder for primary endpoint.

For the primary objective, the statistical null hypothesis states that the complete response rates at the end of 8-week randomized withdrawal period (i.e., at Week 34) are the same between the two randomized arms. To test this hypothesis, a Cochran–Mantel–Haenszel exact test stratified by the two stratification factors considered for randomization will be performed using the RAS following the intent-to-treat principle.

If the 2-sided p-value is ≤ 0.05 and the odds ratio (LCI699 vs. Placebo) is > 1 , the null hypothesis will be rejected and the complete response rate in the LCI699 arm will be considered higher than that in the placebo arm.

As a supportive analysis to this analysis, 1) an un-stratified Fisher's exact test of the primary endpoint will also be performed using RAS; 2) both stratified CMH exact test and un-stratified Fisher's exact test of the primary endpoint will be performed using PPRAS.

2.6 Analysis of the key secondary objective

The key secondary objective is to assess the complete response rate at the end of 24 weeks of dose-titration and treatment with LCI699 in the initial single-arm, open label part of this trial.

The key secondary efficacy variable is the proportion of complete responders at Week 24. A complete responder is defined as an enrolled patient who has $mUFC \leq ULN$ at Week 24 and the dose of LCI699 during Study Period 2 (Weeks 13-24) was not increased above the level established at the end of Study Period 1 (Week 12). Dose reductions and temporary dose interruptions for safety reasons do not preclude patients from being complete responder for the key secondary endpoint. Enrolled patients who had missing $mUFC$ assessment at Week 24 will be counted as non-responders for the key secondary endpoint.

For the key secondary objective, the statistical null hypothesis states that the complete response rate at week 24 LCI699 is $\leq 30\%$. The analysis of the key secondary objective will be based on the 2-sided 95% exact confidence interval (Clopper-Pearson method). If the lower bound of this 95% confidence interval is $\geq 30\%$, the null hypothesis will be rejected and the complete response rate will be considered at least 30% after 24 weeks of treatment with LCI699.

The above testing on the key secondary objective will only be carried out if the null hypothesis for the primary objective is rejected. This sequential procedure will ensure preservation of the overall 2-sided type 1 error at 5%.

The above analysis of the key secondary endpoint will be performed using FAS. As supportive analysis, same analysis will be performed using PPFAS.

2.7 Analysis of secondary efficacy objective(s)

For the objectives identified below, the definition of Baseline/Randomization assessment and the method of computing confidence intervals are specified in Section [2.1.1.2](#) and Section [2.1.1.8](#) respectively. Last observation carried forward (LOCF) method will not be used in any of these analyses. For analysis using RAS, result will be presented by randomized treatment assigned (LCI699 or Placebo). For analysis using FAS, results will be presented by LCI699 group, Placebo group, Non-randomized patients, and All patients.

2.7.1 Compare the time-to-last control of mUFC during the randomized withdrawal period between patients randomized to continued LCI699 and placebo

Time-to-last control of mUFC during the randomized withdrawal period is defined as the time (in days) from randomization to the last normal UFC assessment ($mUFC \leq ULN$ based on central laboratory result) within the randomized withdrawal period. It will be defined as censored if the last UFC assessment is normal and event otherwise.

Time-to-last control of mUFC during the randomized withdrawal period will be analyzed using K-M plot and stratified log-rank test. The quartiles of time-to-last control and corresponding two-sided 95% CI will be calculated using Kaplan-Meier methodology. RAS will be used for the analyses.

2.7.2 Assess the complete, partial, and overall response rate at Week 12, Week 24, Week 48, and during the extension period

Proportion of complete responder (enrolled patients with $mUFC \leq ULN$), partial responder (enrolled patients with $mUFC > ULN$ and at least 50% reduction from baseline), and overall responder (enrolled patients with $mUFC \leq ULN$ or have at least 50% reduction from baseline) will be summarized using point estimate for every scheduled visit in core and extension (provided [adequate follow-up](#)) at which UFC is collected. 95% 2-sided CI for response rates will be provided for Week 12, Week 24, Week 48, and the last available assessment of the study. FAS will be used for the analyses.

2.7.3 Assess the change in mUFC during the core and extension period of study

- Change in mUFC from randomization during the randomized withdrawal period only

For the actual value and change in mUFC from randomization during randomized withdrawal period, descriptive summary will be provided. In addition, change of mUFC during the randomized withdrawal period (on the natural log scale) will be analyzed using the repeated measure model below. RAS will be used for the analyses.

Change of mUFC from randomization (log scale) = $\log(mUFC \text{ at randomization}) + \text{randomized treatment} + \text{Week (from randomization)} + \text{randomized treatment} * \text{Week (from randomization)} + \text{error}$

The fixed effect terms randomized treatment and Week (from randomization) will be included as classification variables, and log(mUFC at randomization) will be included as a continuous covariate. The Unstructured covariance structure will be assumed as the covariance structure within patients. In case the model does not converge, Toeplitz or AR(1) covariance structure will then be used. The least square means for mUFC change from randomization (log scale) for each randomized treatment as well as their difference will be transformed back into the original scale as a % change from baseline. The corresponding 95% two-sided CIs will be provided.

Per study design (see Section [1.2.3](#)), if the mUFC increases to $> 1.5 \times \text{ULN}$, and at least 2 individual urine samples show $\text{UFC} > 1.5 \times \text{ULN}$ at a single visit during the randomized withdrawal period, the patients must be discontinued and thus will have missing mUFC for the remaining visits within randomized withdrawal period. Given this missing mechanism depends only on the observed mUFC value and thus is Missing at Random (MAR), it is valid to analyze such missing data using the proposed mixed effect model.

- Change in mUFC from baseline during the overall study period

For the actual value and change in mUFC from baseline, descriptive summary will be provided for every scheduled visit in core and extension (provided [adequate follow-up](#)) at which UFC is collected. In addition, a repeated measure model (see below) will also be used to estimate the effect of LCI699 treatment on the change from baseline in mUFC (on the natural log scale) during the core study period. FAS will be used for the analysis. Given the goal of this analysis is not to estimate the difference between randomized treatments, for patients randomized to Placebo, only data before first dose of randomized treatment will be included to enable a straightforward interpretation of result as the effect of LCI699 at ‘optimal’ dose determined by UFC and safety data from individual patient.

Change of mUFC from baseline (log scale) = log (baseline mUFC) + Week (from baseline) + error

The fixed effect terms Week (from baseline) will be included as classification variables, and log(baseline mUFC) will be included as a continuous covariate. Since all data analyzed are on individual ‘optimal’ dose of LCI699 treatment, no treatment/dose indicator is needed in the model. The Unstructured covariance structure will be assumed as the covariance structure within patients. In case the model does not converge, Toeplitz or AR(1) covariance structure will then be used. The adjusted mean change from baseline on the log scale will be estimated at each available post-baseline visit using least-squared means and transformed back into the original scale as a % change from baseline. The corresponding 95% CIs will be provided.

2.7.4 Assess the change in cardiovascular-related metabolic parameters associated with Cushing’s disease during the core and extension period of the study

For each cardiovascular-related metabolic parameter associated with Cushing’s disease (e.g. glucose, HbA1c, fasting lipid profile, SBP, DBP, weight, waist circumference and BMI):

- Change from randomization during randomized withdrawal period only

For the actual value and change from randomization during the randomized withdrawal period, descriptive summary will be provided. In addition, such change during the randomized withdrawal period will be analyzed using repeated measure models described below. RAS will be used for analyses.

$$\begin{aligned} \text{Change from randomization} = & \text{value at randomization} + \text{randomized treatment} + \\ & \text{Week (from randomization)} + \text{randomized treatment} * \text{Week (from randomization)} + \\ & \text{error} \end{aligned}$$

The fixed effect terms randomized treatment and Week (from randomization) will be included as classification variables, and baseline value will be included as a continuous covariate. The Unstructured covariance structure will be assumed as the covariance structure within patients. In case the model does not converge, Toeplitz or AR(1) covariance structure will then be used. The least square means of change from randomization for each randomized treatment as well as their difference at each available time points will be provided. The corresponding 95% two-sided CIs will be provided.

- Change from baseline during the overall study period

For actual and change from baseline, descriptive summary will be provided for every scheduled visit in core and extension (provided [adequate follow-up](#)). In addition, for each individual parameter, the effect of LCI699 treatment on the change from baseline during the core study period will be estimated using repeated measure model described below. FAS will be used for the analysis. Given the goal of this analysis is not to estimate the difference between randomized treatments, for patients randomized to Placebo, only data before first dose of randomized treatment will be included to enable a straightforward interpretation of result as the effect of LCI699 at ‘optimal’ dose determined by UFC and safety data from individual patient.

$$\text{Change from baseline} = \text{baseline value} + \text{Week (from baseline)} + \text{error}$$

The fixed effect terms Week (from baseline) will be included as classification variables, and baseline value will be included as a continuous covariate. Since all data analyzed are on individual ‘optimal’ dose of LCI699 treatment, no treatment/dose indicator is needed in the model. The Unstructured covariance structure will be assumed as the covariance structure within patients. In case the model does not converge, Toeplitz or AR(1) covariance structure will then be used. The adjusted mean change from baseline will be estimated at each available post-baseline visit using least-squared means. The corresponding 95% two-sided CIs will be provided.

2.7.5 Assess the change from baseline in physical features of Cushing's disease by photography

For physical features of Cushing's disease, such as facial rubor, hirsutism, striae, supraclavicular fat pad, dorsal fat pad, proximal muscle wasting (atrophy), central (abdominal) obesity, and ecchymoses (bruises), the change from baseline (captured by a semi-quantitative Likert scale) at visits where photograph is scheduled (Week 12, 24, 34 and 48 as well as Week 72 and EOT extension) will be assessed by shift table. In addition, the proportions of patients having a favorable shift from baseline at post-baseline time points will also be tabulated. FAS will be used for the analyses.

2.7.6 Assess the change from baseline in bone mineral density by DXA scan at Week 48 and the last available assessment

For bone mineral density measured by DXA scan at the lumbar spine and total hip, descriptive summaries of actual and percentage change from baseline at Week 48 and the last available assessment will be provided. FAS will be used for the analysis.

2.7.7 Assess the time-to-escape under treatment of LCI699

Escape is defined as the first loss of control of UFC that meets all of the following criteria:

- prior normalization of UFC has occurred ($mUFC \leq ULN$)
- both the $mUFC$ and at least 2 individual values contributing to that $mUFC$ have to be $>1.5 \times ULN$
- the loss of control of UFC is not related to a dose interruption or dose reduction due to safety reasons
- happens beyond 12-week dose titration period (Study Period 1).

Time-to-escape is defined as the time (in days) between the first UFC assessment with $mUFC \leq ULN$ and either the UFC assessment of escape (event) or last UFC assessment before permanent discontinuation of LCI699 treatment (censored), whichever occurs earlier.

For patients who attained normalization of UFC ($mUFC \leq ULN$), time-to-escape will be analyzed using K-M plot. The quartiles of time to escape and corresponding two-sided 95% confidence interval will be calculated using Kaplan-Meier methodology. In addition, time-to-escape will be summarized using descriptive statistics for patients who escape. FAS will be used for the analysis.

Patients randomized to placebo are not included in this analysis.

2.8 Safety analyses

The assessment of safety will be based mainly on the frequency of adverse events and on the number of laboratory values that fall outside the pre-determined ranges. Other safety data (e.g. electrocardiogram, vital signs, and special tests) will also be presented.

In this study, there are three different study periods during which safety analysis will be performed. Analysis set and treatment group (See Section [2.2](#)) used for each of three study periods are described below:

- For study period up to Week 26: SAS by All patients only.
- For randomized withdrawal period: SASR by LCI699 group and Placebo group
- For overall study period (on LCI699 treatment): SAS by LCI699 group, Placebo group, Non-randomized patients, and All patients.

All safety data available at the data base lock will be included in the safety assessment specified in this SAP. However, cutoff flag will be derived in the analysis data so that potential requests from health authority for safety outputs using core phase data only (up to week 48) can be readily handled.

2.8.1 Adverse events (AEs)

All information obtained on adverse events will be displayed by patient.

The number and percentage of subjects with adverse events will be tabulated by system organ class and preferred term. A subject with multiple adverse events within a body system is only counted once towards the total of this body system if no change in severity.

2.8.1.1 Coding of AEs

Adverse events are coded using the Medical Dictionary for Regulatory Activities (MedDRA) terminology. The latest version available at the time of database lock will be used.

2.8.1.2 Grading of AEs

AEs will be graded according to the NCI Common Terminology Criteria for Adverse Events (CTCAE) version 4.03.

The CTCAE represents a comprehensive grading system for reporting acute and late effects of cancer treatments. CTCAE v4.03 is graded by definition a 5-point scale generally corresponding to clinical severity (mild, moderate, severe, life-threatening, and death). This grading system inherently places a value on the importance of an event although there is not necessarily proportionality among grades (a Grade 2 is not necessarily twice as bad as a Grade 1).

For adverse events for which CTCAE grades are not available, grades 1 – 4 corresponding to the severity of mild, moderate, severe, and life-threatening will be used. CTCAE grade 5 (death) is not used in this study; rather, this information will be collected on the “End of Treatment” or “Study evaluation completion” eCRF pages

2.8.1.3 General rules for AE reporting

AE summaries will include all AEs starting on or after study day 1 (i.e. on or after the day of the first administration of study drug) but no later than AE summary cutoff defined in Section [2.1.1.9](#). All AEs before data cutoff date will be listed, including those that start before study day 1. AEs starting prior to study day 1 will be identifiable based on the AE start date displayed in the listings. [_AE_summary_cutoff](#)

AEs will be summarized by presenting the number and percentage of patients having at least one AE, and having at least one AE in each body system/primary system organ class and for each preferred term using the most current MedDRA coding available prior to database lock. A subject with multiple occurrences of an AE will be counted only once in the AE category.

Separate AE summaries will be presented by primary system organ class, preferred term, and maximum CTC grade. A patient with multiple CTC grades for an AE will be summarized under the maximum CTC grade recorded for the event. In the summaries presented by grade, all AEs will be pooled regardless of whether they are CTC gradable or not. AEs with missing CTC grade will be summarized under “missing”.

The frequency of CTC grades 3 and 4 AEs will be summarized separately.

Any information collected (e.g. CTC grades, relationship to study drug, action taken etc.) will be listed as appropriate.

2.8.1.4 AE summaries

The following adverse event summaries will be produced:

- Adverse events, regardless of study drug relationship, by primary system organ class, preferred term and maximum CTC grade
- Adverse events, suspected to be study drug related, by primary system organ class, preferred term and maximum CTC grade
- Adverse events, regardless of study drug relationship, by preferred term
- Adverse events, suspected to be study drug related, by preferred term
- All deaths, by primary system organ class and preferred term
- Serious adverse events, regardless of study drug relationship, by primary system organ class, preferred term and maximum CTC grade
- Serious adverse events, suspected to be study drug related, by primary system organ class, preferred term and maximum CTC grade
- Serious adverse events, regardless of study drug relationship, by preferred term and treatment
- Adverse events leading to study drug discontinuation, regardless of study drug relationship, by primary system organ class and preferred term

- Adverse events requiring dose adjustment or study-drug interruption, regardless of study drug relationship, by primary system organ class and preferred term
- Adverse events requiring additional therapy, regardless of study drug relationship, by primary system organ class and preferred term
- Serious adverse events leading to study drug discontinuation, regardless of study drug relationship, by primary system organ class and preferred term
- Adverse events of Special Interest, regardless of study drug relationship, by category and preferred term
- Adverse events of Special Interest with suspected study drug relationship, by category and preferred term
- Adverse events of special interest leading to study drug discontinuation, regardless of study drug relationship, by category, and preferred term
- Adverse events of special interest requiring dose adjustment or interruption, regardless of study drug relationship, by category, and preferred term
- On-treatment deaths and serious adverse events by system organ class and preferred term
- Non-serious adverse events (threshold = 5%) by system organ class and preferred term

2.8.1.5 Adverse events of special interest / grouping of AEs

Groupings of AEs of special interest consist of adverse events for which there is a specific clinical interest in connection with LCI699 treatment (i.e. where LCI699 may influence a common mechanism of action responsible for triggering them) or adverse events which are similar in nature (although not identical).

An Excel file with the exact composition of the adverse events groupings is available in the CREDI folder “/CREDI Projects/L/LCI699C/Administrative files/CIS (Clinical Information Sciences)/Biostatistics” which is to be used to map reported adverse events to the adverse events groupings. This file is updated periodically after MedDRA update and/or review of accumulating trial data.

2.8.2 Deaths

Deaths will be reported, by primary system organ class and preferred term. Patients will not be followed-up for survival in this study, with patients considered to be off-treatment 30 days after treatment discontinuation.

2.8.3 Laboratory data

For analyzing laboratory results, data from all sources (central and local laboratories) will be combined. The summaries will include all laboratory assessments up to data cut-off date. All laboratory assessments will be listed. Results will be reported for each visit at which collected in the Core, Follow-up, and Extension Periods through the last study visit.

Eligible laboratory data will be classified into CTC grades according to NCI CTCAE v4.03. In the unlikely case a local laboratory normal range overlaps into the higher (i.e. non-zero) CTC grade, if the laboratory value is within local normal limits it will be assigned a CTC grade of zero.

- For study period up to Week 26 and overall study period

The following summaries will be produced for the laboratory data (by laboratory parameter):

- Number and percentage of patients with worst post-baseline CTC grade (regardless of the baseline status). Each patient will be counted only for the worst grade observed post-baseline
- Number and percentage of patients meeting categorical liver function test criteria, including Hy's Law criteria for liver injury (ALT or AST > 3 x ULN and TBIL ≥ 2 x ULN and ALP < 2 x ULN). Each patient will be counted only for the worst grade observed post-baseline
- Shift tables from baseline to the worst post-baseline value and from baseline to the last post-baseline values using CTC grades will be produced for hematology, biochemistry, and urine laboratory parameters with CTC grades
- For laboratory parameters in which CTC grades are not defined, shift tables from baseline to the worst post-baseline value and from baseline to the last post-baseline values will be produced using the low/normal/high classifications based on laboratory reference ranges
- Shift tables of fasting blood glucose and HbA1c from baseline to the highest post-baseline value and from baseline to the last post-baseline values using the ADA (2010) classification.
- Selected laboratory data will be also be displayed by presenting summary statistics of change from baseline value.

- For randomized withdrawal period

After replacing 'baseline' with 'randomization' and 'post-baseline' with 'post-randomization', the above description for study period up to Week 26 and overall study period will applied.

The following listings will be produced for the laboratory data:

- Listing of all laboratory data with values flagged to show the corresponding CTC grades and/or the classifications relative to the laboratory reference ranges
- Listing of notable laboratory abnormalities (i.e. CTC grade 3 or 4 laboratory toxicities).

2.8.4 Other safety data

2.8.4.1 ECG and cardiac imaging data

Fridericia's formula (QTcF) will be used to calculate the heart rate-corrected QT interval (ms) based on the heart rate (HR bpm) and QT (ms) for centrally-read data. These calculated QTc values will be used in ECG summary tables and listings. The formula is as follows:

$$QTcF \text{ (in ms)} = QT / (RR)^{1/3}$$

The following analyses will be performed for variables PR, QRS, QT interval, heart rate, and QTcF (Fridericia's):

- For study period up to Week 26 and overall study period: summary statistics at baseline, all post-baseline time points as well as corresponding change from baseline.
- For randomized withdrawal period: summary statistics at randomization, all time points within randomized withdrawal period as well as corresponding changes from randomization
- Listing of ECG data including all scheduled and unscheduled time points

No imputation of missing data will be performed.

The number and percentage of patients with notable ECG interval values will be summarized.

- For study period up to Week 26 and overall study period

A newly occurring ECG finding is defined as an notable finding at post-baseline that is not present at baseline.

The notable criteria for PR are

- > 200 ms post-baseline and ≤ 200 ms at baseline
- Increase $> 25\%$ compared to baseline to a post-baseline value > 200 ms

The notable criteria for QRS are

- > 110 ms post-baseline and ≤ 110 ms at baseline
- Increase $> 25\%$ compared to baseline to a post-baseline value > 110 ms

The notable criteria for QT and QTcF are

- > 450 ms at any post-baseline and ≤ 450 ms at baseline
- > 480 ms at any post-baseline and ≤ 480 ms at baseline
- > 500 ms at any post-baseline and ≤ 500 ms at baseline
- an increase from baseline > 30 ms at any post-baseline
- an increase from baseline > 60 ms at any post-baseline

The percentage of patients having notable ECG interval values is based on the total number of patients who are at risk for a specific category. For new abnormality post-baseline values, this is the number of patients with both baseline and post-baseline evaluations who are normal at baseline. For abnormal change from baseline, this is the number of patients with both baseline and post baseline evaluations. A subject with multiple occurrences of a newly occurring abnormality is counted only once per abnormality.

- For randomized withdrawal period

After replacing 'baseline' with 'randomization' and 'post-baseline' with 'post-randomization', the above description for study period up to Week 26 and overall study period will applied.

All ECG data will be listed. If corrected QTcF are provided (providing them is optional), they will be listed in addition to the values derived using the formulae above. (In the summaries

only the derived values will be used.) The patients with notable ECG interval values will be listed and the corresponding notable values will be flagged in the listings.

2.8.4.2 Vital signs

All vital signs data (weight (kg), body temperature ($^{\circ}\text{C}$), pulse rate (beats per minute), and systolic/diastolic blood pressure (mmHg)) will be listed by patient and visit/time, and abnormalities will be flagged on listing. Shift table based on notable value will be provided for vital signs. For blood pressure, triplicate measurements are planned when assessed, and the mean of available values will be taken for analysis purposes. Change over time from either baseline (for study period up to Week 26 and overall study period) or randomization (for randomized withdrawal period) will be summarized.

- For study period up to Week 26 and overall study period

The criteria for clinically notable abnormalities are defined as follows:

7.5.3.1 Clinically notable elevated values

- Systolic BP: ≥ 180 mmHg and an increase ≥ 20 mmHg from baseline
- Diastolic BP: ≥ 105 mmHg and an increase ≥ 15 mmHg from baseline.
- Pulse: ≥ 120 bpm with increase from baseline ≥ 15 bpm
- Weight: Increase from baseline of $\geq 10\%$
- Temperature: Increase from baseline of $\geq 39.1^{\circ}\text{C}$

7.5.3.2 Clinically notable below normal values

- Systolic BP: ≤ 90 mmHg and a decrease ≥ 20 mmHg from baseline
- Diastolic BP: ≤ 50 mmHg and a decrease ≥ 15 mmHg from baseline
- Pulse: ≤ 50 bpm with decrease from baseline ≥ 15 bpm
- Weight: Decrease from baseline of $\geq 5\%$

- For randomized withdrawal period

After replacing 'baseline' with 'randomization', the above definition for study period up to Week 26 and overall study period will applied.

2.8.4.3 Tumor volume

For tumor volume evaluated by MRI (or CT) scanning in patients with detectable tumor volume at baseline, descriptive summary of actual tumor volumes as well as its change from baseline will be provided at visits where evaluation is scheduled (Week 24, 48, and 72). For patients without detectable tumor volume at baseline, incidence of new tumor during the study will be assessed. For MRI (or CT) images that are not interpretable for tumor volume,

the longest dimension (in mm) will be listed instead. Due to evaluation schedule, this analysis will only be performed for the overall study period.

2.9 Pharmacokinetic endpoints

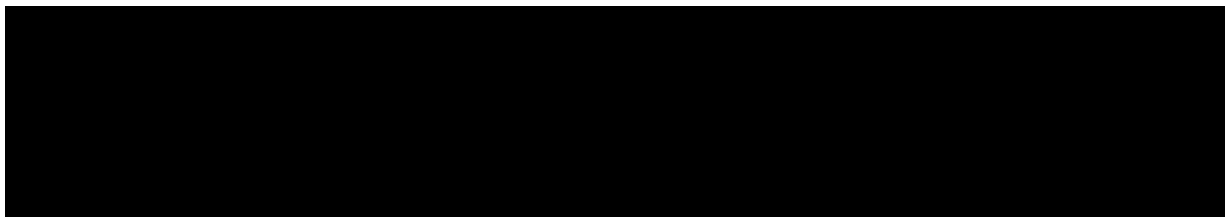
Plasma concentrations of LCI699 will be expressed in ng/mL. All missing concentrations will be labeled as such in the concentration data listings. Concentrations below the lower limit of quantification (LLOQ) will be treated as zero in summary statistics and for the calculation of pharmacokinetic parameters. Alternative methods for handling BLQ may be considered if those reach 5% of the available data for the population PK model.

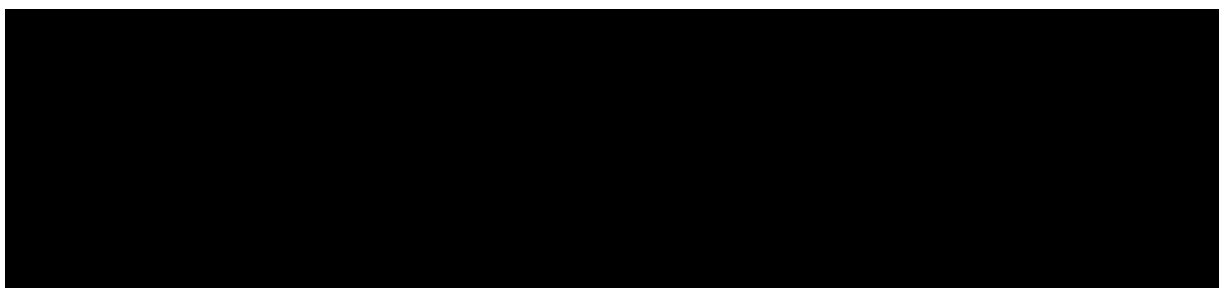
PK observations with missing concentration, missing dose, missing elapsed time, elapsed time outside of acceptable window, or from patients who vomited within 4 hours after dose administration will be excluded from PK statistical summary, covariate analysis, and PK/PD modeling. The acceptable window for Ctrough is pre-dose samples taken within 0.5 h before dose administration, and within 9h to 15h after the previous dose. Analysis will be based on the incident dose (defined as the last dose prior to the PK sample) and use pharmacokinetic analysis set (PAS).

2.9.1 Secondary PK objective - PK exposures of LCI699

As sparse pharmacokinetic sampling is performed in this study, traditional non-compartmental analysis will not be performed to calculate pharmacokinetic parameters. Plasma concentration data of LCI699 will be listed by subject, visit, incident dose and nominal sampling times. Descriptive statistics (n, arithmetic mean, SD, CV% and CV% geo-mean, geometric mean, median, maximum, minimum) of plasma concentrations will be provided by incident dose, visit and nominal sampling times. Graphical depiction (mean and individual) for LCI699 concentrations and/or profiles (if applicable) during the course of the study will be performed by incident dose and visit.

For China patient only, extensive pharmacokinetic sampling will be performed. PK parameters will be calculated from individual plasma concentration versus time profiles using non-compartmental analysis (Phoenix WinNonlin v. 6.2). PK parameters (including AUClast, AUCinf, Cmax, Tmax, T1/2, Cl/F, Vz/F) for LCI699 from these extensive sampling profiles will be summarized using descriptive statistics. For Tmax, only n, minimum, median, and maximum will be presented. Listing of PK concentrations and PK parameters (when applicable) will be provided by patient and visit.





2.11 Patient-reported outcomes

The Cushing's QoL score (both the total score and two subscale scores) is identified as the primary PRO variable of interest. EQ-5D-5L utility index and visual analogue scale (VAS) scores, and Beck Depression Inventory-II (BDI-II) total score, are identified as secondary PRO variables of interest.

2.11.1 CushingQoL

The CushingQoL is a valid and reliable disease-specific QoL questionnaire which assesses health-related quality of life (HRQoL) in patients with Cushing's syndrome and has been validated in patients with Cushing's disease (Webb SM, 2008; Nelson L., 2013). The CushingQoL consists of questions reflecting dimensions of HRQoL related to physical aspects (e.g. 'I bruise easily'), psychological aspects (e.g. 'I am more irritable, I have sudden mood swings and angry outbursts'), and social aspects (e.g. 'I have had to give up my social or leisure activities due to my illness').

The questionnaire consists of 12 items measured on a five point Likert-type scale assessing how often ('always' to 'never') or how much ('very much' to 'not at all') each item has been related to the patient's Cushing's disease in the previous 4 weeks. Scoring of each item ranges from 1 ('Always' or 'Very much') to 5 ('Never' or 'not at all'). For the purpose of the clinical trial, the recall period has been modified to 'within the past week' in order to be more sensitive to the changes in patient HRQoL, specifically during the randomized withdrawal period, where it is believed that changes in Cushing's disease symptoms will occur rapidly once patients withdrawal from LCI699 treatment

Patients who complete 9 or more items at an assessment are considered evaluable for that visit. Standardized scores can be calculated as follows:

The raw score is calculated by summing the individual item scores prior to being standardized so that the total score ranges from 0 to 100, with a lower score indicating a greater impact on HRQoL. The following formula can be used to calculate the total score:

$$[(X - L) / (H - L)] * 100$$

Where X is the total score of the subscale of interest, L is the lowest possible score of the subscale, and H is the highest possible score for the subscale.

Recent research recommends that a 2 sub-scale scoring (namely, a *psychosocial issues scale* and a *physical problems scale*) provides the optimal interpretation, rather than the total score. (Tiemensma J. 2016). The CushingQoL 2 sub-scale scoring of each subscale is exactly the same as the total score. The items which comprise each subscale are shown in the table below:

Table 2-4 Items for the two subscales of CushingQoL

Psychosocial issues subscale	Physical problems subscale
2 <i>I have pain that keeps me from leading a normal life</i>	1 <i>I have trouble sleeping</i>
5 <i>I am more irritable, I have sudden mood swings and angry outbursts</i>	3 <i>My wounds take a long time to heal</i>
6 <i>I have less self-confidence, I feel more insecure</i>	4 <i>I bruise easily</i>
7 <i>I am worried about the changes in my physical appearance due to my illness</i>	
8 <i>I feel less like going out or seeing relatives or friends</i>	
9 <i>I had to give up my social or leisure activities due to my illness</i>	
10 <i>My illness affects my everyday activities such as working or studying</i>	
11 <i>It is difficult for me to remember things</i>	
12 <i>I am worried about my health in the future</i>	

Currently a minimal important difference (MID) of 10.1 is defined in the literature based on the distribution method of a 0.5 SD unit change using baseline data, with the original 4 week recall period (Nelson, 2013, Norman, 2003). However, anchor-based methods for calculating MIDs have been referenced as more reliable estimates for interpreting change in PRO scores over time (based on an individual's level of change rather than a group level of change) in PRO guidelines (FDA, 2009). Given the importance of anchor-based MIDs and that the recall period has been modified to 1 week, the MID will be derived by Novartis from data collected in previous clinical trial(s) using the CushingQoL, as possible, and analyses of the present clinical trial data will then be performed for confirmatory purposes for this anchor-based MID (as possible). The proportion of patients whose CushingQoL standardized score change from baseline by at least the newly derived MID and the associated 95% CI will be summarized by study period and group (specifically for study period 3 analyses).

2.11.2 BDI-II

The BDI-II is a patient-reported instrument developed to measure the severity of depression in adults and adolescents aged 13 years and older. The BDI-II is designed to be completed by the patient on paper and takes approximately five minutes to complete. The BDI-II comprises 21-items assessing the common cognitive symptoms of depression over the previous two weeks. Items are rated on a four-point severity scale of 0 ('not at all') to 3 ('extreme' form of each symptom) with differing response options for each item. A global score ranging from 0 to 63 is calculated with a higher score representing a greater level of depression.

The following scoring guidelines for interpretation of the BDI-II have been suggested (Smarr, 2011):

Minimal range = 0-13

Mild depression = 14-19

Moderate depression = 20-28

Severe depression = 29-63

A minimal clinically important difference for improvement in BDI-II scores has been reported as a 17.5% reduction in scores from baseline; however, this is dependent on baseline severity (Button Ks, 2015)

2.11.3 EQ-5D-5L

Health status will be assessed using the EQ-5D-5L. The EQ-5D-5L has two components, the EQ-5D-5L descriptive system and the EQ-5D-5L visual analog scale (VAS) (Herdman, M., 2011, The EuroQoL Group, 2013). The EQ-5D-5L descriptive system comprises the following five dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension has five levels: 'no problems', 'slight problems', 'moderate problems', 'severe problems' and 'unable to do activity or extreme problems'. A utility index can be computed from the EQ-5D-5L descriptive system with utility scores ranging from -0.281 (worst imaginable health state) to 1 (best imaginable health state), with -0.281 representing an "unconscious" health state. The EQ-5D-5L VAS records the subject's self-rated health state on a 100-point vertical VAS (0=worst imaginable health state; 100=best imaginable health state) (The EuroQoL Group, 1990). A score difference of 0.037-0.069 for the EQ-5D-5L utility score will be used as MID (McClure, 2017). A score difference of 7 for the EQ-5D-5L VAS will be used as MID estimates for the EQ-5D-5L (Pickard, 2007).

The relevant scoring algorithm will be applied to patient-level data in order to derive EQ-5D - 5L index-based scores. The United Kingdom Measurement and Valuation of Health study value set is generally considered the base case scoring function for the purposes of publication (Dolan 1997). Therefore, the EQ-5D-5L utility scores will be based on United Kingdom values for reporting in the CSR. An example of calculating the utility index from raw scores on the EQ-5D-5L are shown below:

Table 2-5 Illustration of calculating the utility index from raw score on the EQ-5D-5L

	Central estimate	Standard Deviation	Value for health state 23245
Constant	1.000		1.000
Mobility			
slight	0.051	0.004	0.051
moderate	0.063	0.004	
severe	0.212	0.006	
unable	0.275	0.006	
Self-care			
slight	0.057	0.004	0.076
moderate	0.076	0.004	
severe	0.181	0.005	
unable	0.217	0.005	
Usual activities			
slight	0.051	0.004	0.051
moderate	0.067	0.004	
severe	0.174	0.005	
unable	0.190	0.005	
Pain/discomfort			
slight	0.060	0.004	
moderate	0.075	0.005	
severe	0.276	0.007	0.276
extreme	0.341	0.008	
Anxiety/depression			
slight	0.079	0.004	
moderate	0.104	0.005	
severe	0.296	0.007	
extreme	0.301	0.007	0.301
Probability (group 1)	0.397	0.019	$0.397 \times 0.427 + 0.270 \times 0.939 + 0.333 \times 1.635$ $= 0.9675$
Probability (group 2)	0.270	0.018	
Probability (group 3)	0.333	0.018	
Slope (group 1)	0.427	0.031	
Slope (group 2)	0.939	0.067	
Slope (group 3)	1.635	0.017	
The value for health state 23245			$1 - 0.9675 \times (0.051 + 0.076 + 0.051 + 0.276 + 0.301)$ $= 0.270$

2.11.4 Secondary objective - Assess the change in Patient-Reported Outcomes (Health-Related Quality of Life) during the core and extension period of the study

The following general guidelines apply to all analyses:

- Changes in PRO that are either statistically significant (based on two-sided 95% confidence interval with no adjustment for multiplicity) or exceeding the MID will be flagged.
- Graphical display(s) of descriptive results will be provided.
- Similar analyses could be performed in subgroups defined by mUFC responder status (controlled, partial controlled and uncontrolled).

For each of four PRO variables (CushingQoL score (2 subscale and total score), BDI total score, EQ-5D-5L utility index, and EQ-5D-5L VAS score):

- Change from randomization during randomized withdrawal period only

For the actual value and change from randomization during the randomized withdrawal period, descriptive summary will be provided. In addition, such change during the randomized withdrawal period will be analyzed using repeated measure models below. RAS will be used for analyses.

Change from randomization = score at randomization + stratum for randomization + randomized treatment + Week (from randomization) + randomized treatment*Week (from randomization) + error

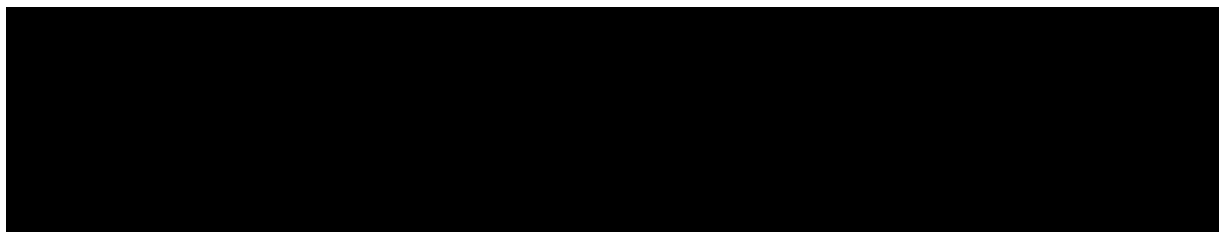
The fixed effect terms randomized treatment, stratum for randomization and Week (from randomization) will be included as classification variables, and baseline value will be included as a continuous covariate. The Unstructured covariance structure will be assumed as the covariance structure within patients. In case the model does not converge, Toeplitz or AR(1) covariance structure will then be used. The least square means of change from randomization for each randomized treatment as well as their difference at each available time points will be provided. The corresponding 95% two-sided CIs will be provided.

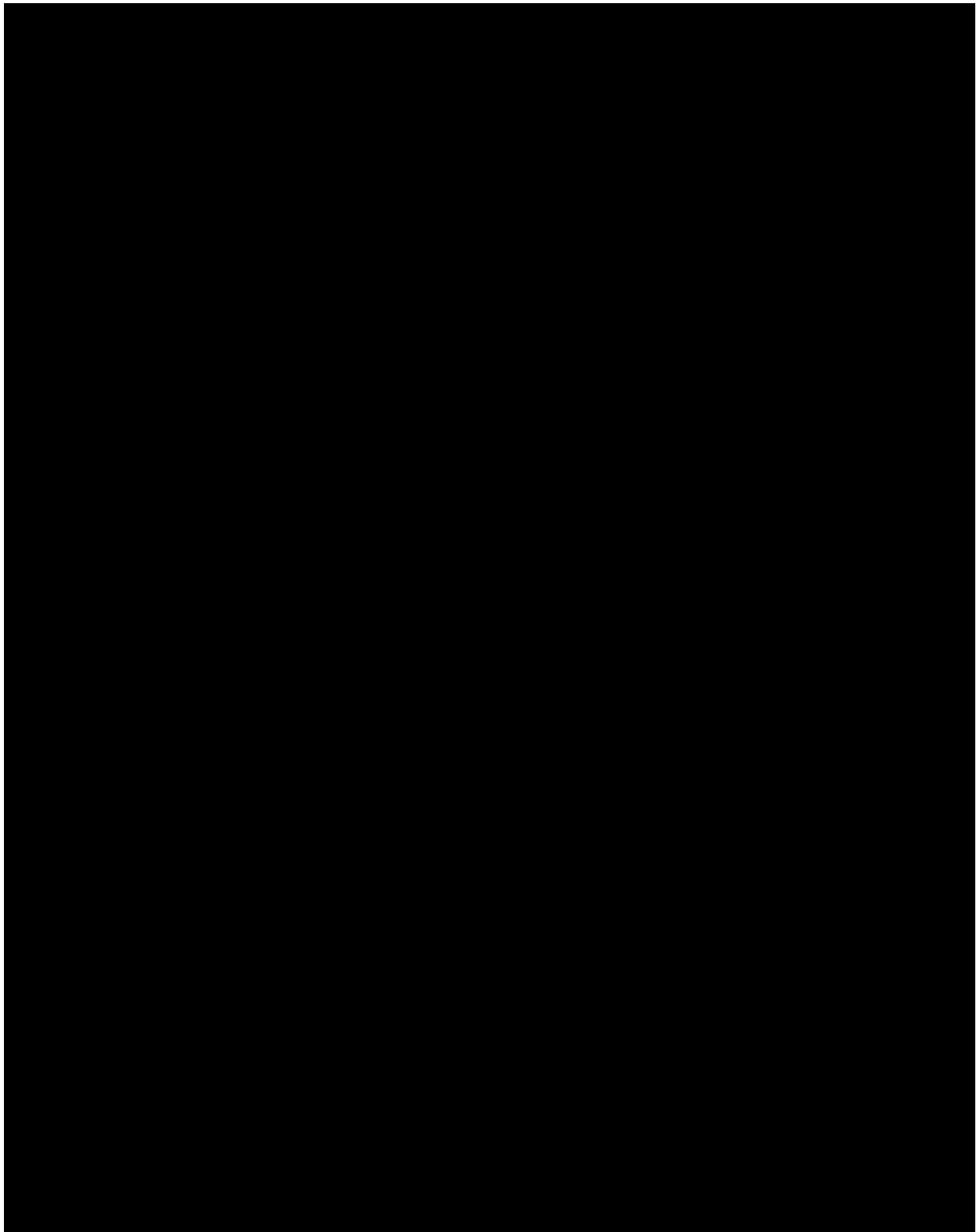
- Change from baseline during the overall study period

For actual value and change from baseline during the core phase of the study, descriptive summary will be provided. In addition, the effect of LCI699 treatment on the change from baseline during the core study period will be analyzed using the repeated measure models below. FAS will be used for the analysis. Given the goal of this analysis is not to estimate the difference between randomized treatments, for patients randomized to Placebo, only data before first dose of randomized treatment will be included to enable a straightforward interpretation of result as the effect of LCI699 at ‘optimal’ dose determined by UFC and safety data from individual patient.

Change from baseline = score at baseline + Week (from baseline) + error

The fixed effect terms Week (from baseline) will be included as classification variables, and baseline value will be included as a continuous covariate. Since all data analyzed are on individual ‘optimal’ dose of LCI699 treatment, no treatment/dose indicator is needed in the model. The Unstructured covariance structure will be assumed as the covariance structure within patients. In case the model does not converge, Toeplitz or AR(1) covariance structure will then be used. The adjusted mean change from baseline will be estimated at each available post-baseline visit using least-squared means. The corresponding 95% two-sided CIs will be provided.





2.14 Interim analysis

The study has no planned interim analysis for efficacy.

3 Sample size calculation

3.1 Sample size justification

Eligible patients will be stratified at randomization according to: LCI699 dose at Week 24 ($\leq 5\text{mg bid}$ vs. $> 5\text{mg bid}$); and history of pituitary irradiation (yes/no). It is estimated that 10 %, 40%, 10% and 40 % of randomized patients respectively will be in the 4 strata defined by two randomization stratification factors ($\leq 5\text{mg b.i.d}$ and Yes; $\leq 5\text{mg b.i.d}$ and No; $> 5\text{mg b.i.d}$ and Yes; and $> 5\text{mg b.i.d}$ and No) based on the following considerations:

- LCI699 dose at week 24 ($\leq 5\text{mg b.i.d.}$ vs. $> 5\text{mg b.i.d.}$): Based on the data from the PoC study, 5mg b.i.d. is estimated to be the median LCI dose at week 24 for this phase 3 trial.
- History of pituitary irradiation (Yes/No): Although supportive data is not available, it is assumed that approximately 20% of randomized patients will have a history of pituitary radiation.
- In the absence of data to expect otherwise, it is assumed that these two stratification factors are independent of each other.

To detect a difference of 40% in complete response rate between 70% in LCI699 arm and 30% in placebo arm (equivalent odds ratio equals 5.444), a sample size of 33 patients per arm will be considered adequate based on a two-sided Cochran-Mantel-Haenszel (CMH) test at the 0.05 level with 87% power. Assuming that 50% of enrolled patients will be eligible for randomization ($\text{mUFC} \leq \text{ULN}$ at the end of the 24 week open-label LCI699 study period), 132 patients need to be enrolled into the study.

3.2 Power consideration for analysis of key secondary objective

The analysis of the key secondary objective will be based on the 2-sided 95% confidence interval constructed using the Clopper-Pearson exact method. If the lower bound of this 95% confidence interval is $\geq 30\%$, the null hypothesis will be rejected and the complete response rate will be considered at least 30% after 24 weeks of treatment with LCI699.

The above testing on the key secondary objective will only be carried out if the null hypothesis for the primary objective is rejected. This sequential procedure will ensure preservation of the overall 2-sided type 1 error at 5%.

Assuming a 50% complete response rate ($mUFC \leq ULN$) at end of 24 weeks of LCI699 treatment, for 132 enrolled patients, there is 99.7% chance that the lower bound of 95% 2-sided CI of observed response rate (based on Clopper-Pearson Exact method) is larger than 30.0%.

4 Change to protocol specified analyses

The time-to-last control of mUFC analysis (section 2.7.1) is changed to be time-to loss of control of mUFC during the RW period.

The rationale for the change was to provide a clear definition of events and censoring which is based on mUFC values and to align with the criteria for discontinuing a patient from randomized treatment due to loss of control.

In the time-to-last control analysis if a patient does not have a normal value during the RW period, they are assigned to have an event at day 1, which is less useful for demonstrating at which time point patients lost control. UFC samples should be collected just prior to the Week 2 visit of the RW period, and so the date of the last collected sample will represent a real observation of exactly when a patient was shown to have lost control.

Time-to-loss of control during the RW period is defined as the time (in days) from randomization to the first evidence of loss of control (defined as mUFC assessment $>1.5 ULN$ based on central laboratory result and at least 2 of the associated individual urine samples show $UFC > 1.5 \times ULN$) within the RW period. A patient without evidence of loss of control will be censored at the date of the last assessment with $mUFC \leq 1.5 ULN$ during the RW period. If a patient discontinues randomized treatment without having a UFC assessment, they will be censored at the date of randomization.

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